

- 3** The matrix $\mathbf{M}$ is given by $\mathbf{M} = \begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, where k is a constant and $k \neq 0$ and $k \neq 1$.
- (a)** The matrix $\mathbf{M}$ represents a sequence of two geometrical transformations. State the type of each transformation, and make clear the order in which they are applied. [2]

The unit square in the x - y plane is transformed by $\mathbf{M}$ onto parallelogram $OPQR$.

- (b) Find, in terms of k , the area of parallelogram $OPQR$ and the matrix which transforms $OPQR$ onto the unit square. [3]

- (c) Show that the line through the origin with gradient $\frac{1}{k-1}$ is invariant under the transformation in the $x-y$ plane represented by $\mathbf{M}$. [3]